

基于 F240 的永磁同步 电机控制系统通信的实现

The Achievement of Communication between PC and Control System Based TMS320F240 for Permanent Magnet Synchronous Motor

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摘 要: 介绍了实现 PC 和 TMS320F240 的串行通信的设计方法, 结合同步电机控制系统, 详细地描述了 TMS320F240 串行通信接口和模块的结构特点, 给出了相关的框图和程序。

关键词: DSP; 串行接口 (SCI); 异步通信控制器; 永磁同步电机

Abstract: The article introduces the design method to carry out the serial communication between PC and TMS320F240, describes the structure characteristic of the serial communication interface and module of TMS320F240 particularly combined with synchronize motor control system, and presents relative frame and program.

Keywords: DSP Serial interface Asynchronous communication controller Permanent Magnet synchronize motor

1、概述

TMS320F240 (以下简称 F240) 是美国 TI (Texas Instruments) 公司特为数字电机控制而推出的一种低价格、高性能 16 位定点 DSP 芯片。它集 DSP 的信号高速处理能力和适用于电机控制的外围电路于一体, 为电机控制系统设计提供了一个理想的解决方案实现形式。在永磁同步电机控制系统的开发过程中, 我们利用 F240 作为电机的微处理器, 在每个一定的控制周期后, 下位机需要将数据送到上位机中进行显示和处理, 同时, 在电机控制中, 同样需要将一些初始数据和调节器的参数实时的进行修改, 这是不能用外接的控制电路端口来控制的, 所以必须应用 F240 芯片中的串行接口完成这些工作。整个监控系统由一台 PC 机和一个以 F240 为核心的电机控制系统组成。电机控制系统完成对电机的反馈控制并采集于电机控制有关的模拟量, 完成双闭环控制后, 将根据 PC 机的指令通过串行通信口将数据发送到 PC 机中进行处理、分析和显示。同时 PC 机还可以通过串行通信改变 F240 程序中转速给定、PI 调节器参数等变量, 发启、停信号等, 实现对控制系统的调试和控制。

利用 Visual Basic 所提供的 MSComm 控件, 可以简单便捷地实现应用程序的串口通信, 甚至创建功能完备、事件驱动的高级通信工具。

1. Summarization

TMS320F240 (called 'F240' for short thereafter) produced by America Texas Instruments Company is the 16 bits DSP COMS chip with lower price and higher performance for digital motor control. This chip integrates the dispose capability with high speed of DSP signal and the periphery circuit used in the motor control. It provides an ideal solution for motor control system. In the exploitation processor of control system of permanent magnet synchronize motor, we use F240 as microprocessor of motor. After every certain control period, the data should be sent to exigency computer from hypogyny computer and displayed and disposed in exigency computer.

At the same time, in motor control, some original data and parameters of control adjustor should be modified in time, which can't be controlled by periphery control circuit port. So these work must be accomplished by serial interface of 'F240 chip. The monitor and control system is composed of the PC and the motor control system in which 'F240 acted as core. The motor control system accomplishes the feedback control on motor and collects analog data concerned motor control. After the double closed loop control is completed, the data

will be sent to PC through serial communication port accord to the instruction of PC, in which they are disposed, analyzed and shown. Simultaneously, the PC can modifies the given rotate speed in 'F240 program and the variable such as parameter of adjustor, sends start-up and stop signal via serial communication, and achieves the debugging and control of control system.

MSComm control widget given by Visual Basic may achieve the serial communication of application program conveniently, even may establish the advanced communication tool with mature function and accident drive.

Compared with the communication procedure programmed by C under the condition of DOS and the serial communication procedure programmed by VC under the condition of Windows, the programmers MSComm widget simply concerned the interface of API function between MSComm widget and the communication drive of Windows when they use MSComm widget of VC to carry out the serial communication. In other words, they simply need to set and monitor the property and events of MSComm widget.

2.F240 Serial Communica-

与DOS环境下用汇编或C等语言编制的通信程序,以及Windows环境下用Visual C编制的串行通信程序相比,利用Visual Basic的MSComm控件实现串行通信时,编程人员只需要关心MSComm控件对Windows通信驱动的API函数的接口。换句话说,只需要设置和监视MSComm控件的属性和事件即可。

2、F240 串行通信接口(SCI)

所有的TMS320C24X器件都包括一个串行通信模块,它支持RS-232和RS-485的工业标准全双工通信模式。串行通信接口的接收器和发送器都是双缓冲的,每一个都有自己独立的使能和中断位,它们可以单独或同时工作于全双工模式。为了保证数据的正确性,串行通信接口对接收的数据进行间断检测、奇偶性、超时和帧错误的检查。通过对一个16位波特率选择寄存器进行编程,可以选择多种不同的波特率,其中最高可达到1.8Mbit/s。

串行通信接口模块的主要特征如下:

- 1) 异步通信格式(NRZ);
- 2) 可编程波特率;
- 3) 可编程数据字长,从1位到8位;
- 4) 1位或2位的可编程停止位;
- 5) 四种错误检测标志位:奇偶性错、超时错、帧错误和间断检测;
- 6) 两种多处理器唤醒模式:空闲线唤醒,地址位唤醒;
- 7) 双工或全双工操作;
- 8) 缓冲接收和发送功能;
- 9) 独立的发送器和接收器使能和中断位。

串行通信接口结构图如图1所示。图中TX.Cntl.Reg和RX.Cntl.Reg是发送器和接收器控制寄存器,用来控制接收使能和中断使能等功能,其实际功能由配置串行通信接口控制寄存器1和2(SCITL1&SCITL2)来实现;TX Buffer寄存器,可获得超过65000种不同的可编程波特率,当系统时钟为10MHz时,可选范围为19.7kbit/s到625kbit/s;SCI Ctrk.Reg.为串行通信接口通信控制寄存器,定义用于SCI的字符格式、协议和串行通信模式。

SCI有两个多处理器协议,空闲线多处理器模式和地址多处理器模式。其中空闲线多处理器模式是在地址字节前特意留出十位或更多的空闲时间来延时,而没有附加的地址/数据位,它在处理包含多于12个字节的数据帧方面比地址位多处理器模式更有效,由于DSP只要通

tion Interface

All of TMS320C24X components have a serial communication module, which sustains the industry standard full-duplex communication pattern. The receiver and the sender of serial communication interface are double-cushion, each of them has respectively enabled and interrupt digit. They can works in full-duplex pattern solely or together. For the sake of the correctness of data, SCI inspects interruptive examination, parity-spouse character, overtimes and frame falseness of received data. Through programming on one 16-bit baud rate elect-register, it may choice multi-different baud rates, among which the largest value is 1.8Mbit/s.

The following is the primary character of SCI module:

- (1) Asynchronous communication format;
- (2) Baud rate programmed;
- (3) The bit of data programmed from 1bit to 8 bits.
- (4) One or two stop bits programmed;
- (5) Four kinds of error inspect symbol bit of error inspect: parity-spouse error, over-time error, frame error and interruptive inspection;
- (6) Two kinds of multi-processor awake pattern: vacancy line awake, address bit awake;
- (7) Duplex or full-duplex operation;
- (8) Cushion take-over and transmission function;
- (9) Absolute enabled and interrupt digit of sender and receiver.

The communication interface of SCI is shown as figure 1. TX.Cntl.Reg and RX.Cntl.Reg is control register of sender and

receiver, which may realize receive and interrupt enable-disable functions.

In fact the function is achieved by SCITL1&SCITL2; TX Buffer is cushion register of sending data, which includes the data to be sent. When data needs to be sent, it may be written into this register by CPU; RX Buffer receives cushion register, which includes data received form communication route, which is read by CPU.

TXSHF Reg. and RCSHF Reg. respectively are transmission and take-over shift register; BR Sel.Reg is a choice register of 16 bit baud rate from which more than 65000 different kinds of baud rate programmed can be get. When the system clock is 10MHz, the scope of baud rate is from 19.7kbit/s to 625kbit/s; SCI Ctrk.Reg is a communication control register of SCI, which defines the character format, protocol and serial communication pattern used for SCI.

SCI has two multi-processor protocols: vacancy route multi-processor pattern and address multi-processor pattern. Vacancy route multi-processor pattern especially sets aside ten bits or more vacancy before address bit for delay, yet it hasn't additional address/data bit. It is more efficiently than address multi-processor in processing data frame over 12 bytes. Because DSP may achieve vacancy signal before address byte easily through corresponding register setting, this pattern should be used in non-multi-computer serial communication in typical DSP. While the address multi-processor joins one bit (one means address frame, yet zero means data frame) in each byte in order to detach address from data, this pattern is used in less 11 bytes data frame typically.

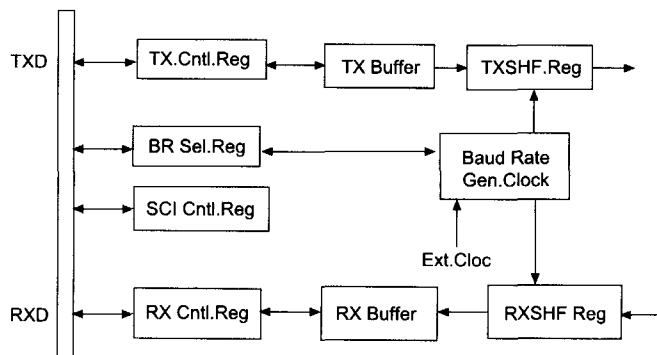


图1 TMS320F240的串行通信接口模块结构简图

Figure1. The structure of TMS320F240 SCI module

过相应的寄存器的设置很容易实现地址字节前的空闲信号,所以该模式应用于典型的DSP之间非多机串行通信。地址位多处理器模式则在每个字节中加入一个位值(1表示地址帧,0表示数据帧)来将地址同数据分开,这种模式典型地用于少于11个字节的数据帧。

3、串行通信实现方法及协议

在通信过程中,为实现监测和控制所需的实时性,应用了RS232C双工通信。由于电机控制程序始终在运行中,不能采用通信过程来中断它,所以只能使用程序查询方式控制DSP串行端口发出实时数据。一方面,根据微机端的指令,DSP持续发送所选定的信号量,另一方面,微机可由人为控制向F240发出控制指令和数据,在F240端的程序中,接收的数据作为“选项”进入“虚拟菜单”,改变F240中相应的程序设置,整个通信的控制由微机中由Visual Basic建立的控制平台来完成。

在Windows 95/98下实现串行通信的方法很多,一般是使用编程语言来实现,这种方法调用内部的端口读写函数或模块来完成串行口的收发控制,能够生成独立运行的小规模软件。另外一种方式是在专用的软件包内调用其串行口通信模块,如LabVIEW,在其软件环境下完成通信和控制,这种方式的缺点是要依赖于规模较大的软件和专用的采样卡运行,所以我们首先采用了第一种方法,在VB平台下完成系统编程。

用VB6.0实现串行通信有2种方法。一是使用串行通信控件(MSComm32.OCX);二是使用Windows95应用编程接口(API)。VB中的MSComm.VBX用户通信控件提供了通过串口发送和接收数据的串行通信能力,不但包括了全部Windows API中关于串行通信的16个函数所完成的功能,而且开拓了更多的对象属性,为串行通信控件提供了标准的事件处理函数、过程,并通过属性的方法提供了串行通信参数的设置,而API提供了完备的应用程序接口函数和中断方式的通信设备驱动程序(COMM.DRV),这两种方法都能访问微机的标准端口,但使用MSComm32控件相对简单。

PC和F240应采用相同的通信协议格式才能实现正确的通信,通信格式定义为:8个数,数据位,禁止奇偶校验,波特率为9600bit/s,停止位为1位。

3.The Way and protocol to accomplish Serial Communication Accomplish

In order to accomplish the real time of inspect and control, RS232 duplex communication is applied to communication processing. Because motor control system is always running, it can't be interrupted by communication procession. The program query mode is used for controlling DSP serial port sending real time data. On the one hand, DSP sends signal chosen continuously according to the instruction of PC port; On the other hand, PC could be controlled artificially to send control instruction and data to 'F240. In 'F240's program, the data received enters "dummy menu" as "option" and modifies the corresponding program setting 'F240's. The control flat roof built by Visual Basic in PC accomplishes all the communication control.

There are many ways to carry out the serial communication under the condition of WIN95/98 Mac OS. Currently it can be carry out by using programme language, which may accomplish receive and send control of serial port through transferring inner port read-write function or module, and may create in miniature software running independently. The other way is transferring the communication module of serial port in appropriate software bundle, such as LabVIEW. Under the software condition of LabVIEW, the communication and the control are accomplished. The shortage of this way is that it's running depends on the biggish software and special sampling card. Therefore we adopt the first way firstly to finish the program on system under VB flat proof.

There are two ways to carry out the serial communication using VB6.0. One is using the serial communication control (MSComm32.OCX); the other is using Windows95 application programming interface (API).

MCSomm.VBX user communication control of VB supplies the serial communication ability of receiving and sending data via the serial port. It includes all functions of sixteen functions about the serial communication in Windows AP. Moreover, it explores more object properties, which of-

fers normal functions and processions of event handling to the serial communication controls and the setting of the parameters of the serial communication by the way of properties. API offers the self-contained interface functions of the application programs and the driver for communication facilities by the way of interrupt (COMM DRV). In the two ways mentioned above, the former is easier to access the standard port of MC.

In order to actualize accurate communication, PC and 'F240 should adopt same communication format. The following is the definition of the communication format: 8 data bit, prohibiting parity checkout, the 9600bit/s baud rate is, one stop bit.

The connection pattern from PC to F240 is shown as figure2.

4.The design of MC Communication Program

MSComm32.OCX has two different ways to solve all kinds of the exploitation and design problems of communication software:

(1) Query mode: Read CommEvent's value and dispose mistake or accident happened. This way is usually adopted in easy design of application program.

(2) Event driven: It is able to dispose many correlative accident of communication, which analogize to the window callback function of C/C++. For example, when the circuitry data gets to the root end or the CD line and RST signal state varies, they desired to be tracked and disposed by relevant accidents which is achieved by using OnComm accident. It also includes the inspection and disposal of communication mistakes and so on. The CommEvent value returns the latest communication accident or the false data code. The key of programming communication program using communication control is understands the attribute setting of communication control correctly. The MSComm.VBX supplies 27kinds of attributes about the communication control. The important ones among them is shown below:

- Comport: Setting or returning the communication port number;
- Setting: Setting or returning the data

F240 与 PC 机的连接方式如图 2 所示。

4、微机通信程序设计

MSComm32.OCX 有两种不同的方式来解决各类通信软件的开发和设计问题:

(1) 查询方式: 读取 CommEvent 的值并处理所发生的错误或事件, 通常简单的应用程序设计可采用这种办法。

(2) 事件驱动: 它与 C/C++ 中的窗口回调函数类似, 能及时地处理许多通信中的相关事件, 例如: 当线路数据到达本端或 CD 线和 RST 信号线状态发生变化时, 要求我们使用相应的事件来跟踪和处理, 该控件是使用 OnComm 事件来实现的, 它也包括检测和处理通信错误等方面的问题。由 CommEvent 值返回最近的通信事件或错误的代码。

利用通信控件编制通信程序, 关键是准确理解设置通信控件的属性, MSComm.VBX 提供了 27 个关于通信控件方面的属性, 其中重要的有:

- CommPort: 设置或返回通信端口编号
- Setting: 设置或返回字符串形式的数据通信格式: 波特率、校验、数据位、停止位
- PortOpen: 设置或返回通信口状态 (包括打开或关闭一个通信口)

为适应实时监控的要求, 并提高程序的运行效率, 本系统采用事件驱动方式开发 MSComm 控件的应用, 在程序中如下设置:

“设置当前通信串口为 COMM2 口”

MSComm1.Commport=2

“设备串口传输速率为 9600bps, 数据位 8 位, 无校验, 一位停止位

MSComm1.Setting=“2400,n,8,1”

“以二进制形式从输入缓冲区读取数据”

MSComm.InputMode=1

“接收到数据引发 OnComm 事件”

MSComm1.Rthreshold=1

“一次读出输入缓冲区中的一个数据”

MSComm1.InputLen=1

“打开端口”

MSComm.PortOpen=True

Private Sub Command1_Click()

.....

在 PC 机的通信程序界面中有 1 个通信控件、几个命令按钮控件。通信控件用于访问串口, 发送和接收数据; 命令按钮控件用于发送命令和参数设置。数据接收采用事件驱动的通信方式, 由 OnComm 事件以接收数据并处理, 选

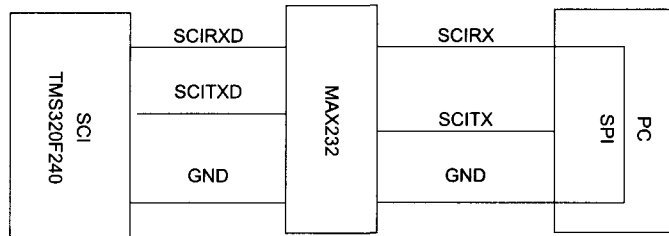


图 2 PC 机与 F240 通信连接图

Figure2. The Communication Connection Figure of PC and F240

communication format in form of character string format: baud rate, verify, data bit, stop bit.

● PortOpen: Setting or returning the communication state (It includes opening or closing one communication port.)

In order to adapt to the demand of real time inspection and control and to enhance the running efficiency of program, this system adopts event-driven mode to exploit the application of MDComm control. The setting in program is shown as following:

“Setting currently communication serial port as COMM2 port.

MSComm1.Commport=2

“Setting the transfer rate of serial port as 9600bps, the data bit as 8 bits, no verify, one stop bit.

MSComm1.Setting=“2400,n,8,1”

“Reading data from the import buffer in binary form.

MSComm.InputMode=1

“Receiving the data solicitation OnComm event.

MSComm1.Rthreshold=1

Reading one data from the import buffer each time.

MSComm1.InputLen=1

“Opening the port.

MSComm.PortOpen=True

Private Sub Command1_Click()

There is one communication control and several command button controls in the communication program interface of PC. The communication control is used to access the serial port, send and receive data; the command button control is used to send commands and set parameters. Receiving data adopts event-driven communication mode. In this way, the OnComm event receives and disposes data, and chooses the observation variable, then the serial port sends ASCII code appointed; the electricity speed given is controlled by the scroll bar, which sends out instruction word and two given data words; the parameter of PI adjuster is set by the enacted interface, which sends out instruction word and one given

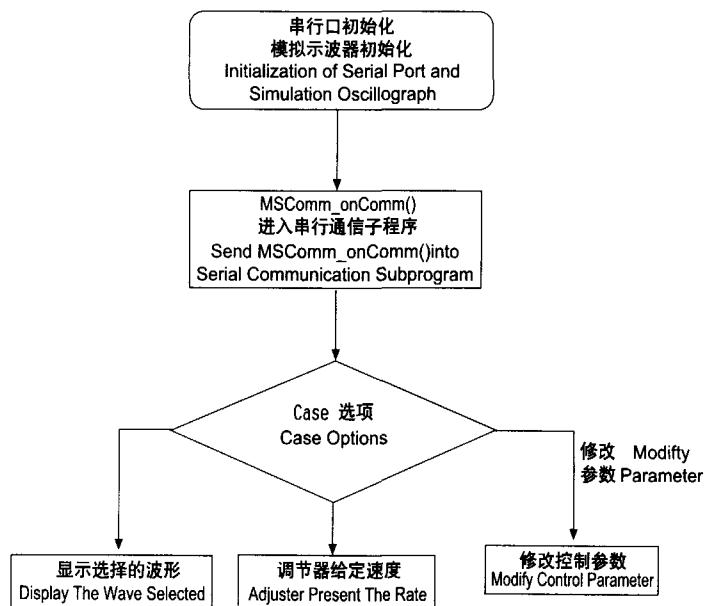


图 3 VB 主程序框图

Figure3. The Structure Frame of VB Main Program

择好观测量后,由串口发出指定的 ASCII 码;电机速度给定由滚动条控制,首先发出指令字,再发出两个给定数据字,PI 调节器参数给定由参数设定界面中设置,首先发出指令字,再发出一个给定数据字。

5、DSP 通信程序设计

TMS320F240 的可编程的 SCI 模块支持不归零格式的 CPU 和其它异步外设之间的数字通信,模块包括了接受缓冲区 (RX) 和发送缓冲区 (TX) 两个具有独立的使能和可屏蔽位的数据缓冲区,可编程波特率发生器,数据区反映射控制寄存器和状态寄存器,SCI 的接收、发送缓冲区数据是 8 位的。RX、TX、clk 的使能由 SCI 控制寄存器 1 (SCICTL1) 控制,SCI 中断的屏蔽由 SCI 控制寄存器 2 (SCICTL2) 控制,波特率由 SCIHBAUD 和 SCILBAUD 设定。SCI 端口控制寄存器 SCIPC2 控制了 SCI 的接收和发送端的 I/O 通信,这些设置在 SCI 的初始化里完成。

SCI initialization

```
LDP DP_PFI
SPLK #00010111b, SCICCR ;
one stop bit ,no partity, 8 bits
SPLK #0013h, SCICTL1 ;
enable RX, TX,clk
SPLK #0000h, SCICTL2 ;
disable SCI interrupts
SPLK #0000h, SCIHBAUD ; MSB
SPLK #0040h, SCILBAUD :
LSB 9600 Baud for sysclk 10Mhz
SPLK #0022h, SCIPC2 : I/O setting
SPLK #0033h, SCICTL1
; end initialization
```

DSP 的通信程序的结构和微机部分程序是一一对应的,所以包含了很多的出口可能性,如观测量的选择、参数设置、速度给定等。为了使汇编程序得到优化,降低多重选择和程序中断造成的不稳定,我们采用了虚拟菜单的方法,如图 4 所示:

虚拟菜单的输入控制即为串行口的传输字,在收到一个传输字后,首先判断其是属于观察值给定命令还是参数设定命令。观察值给定命令对应于主控窗的按钮选项 参数设定命令对应于参数设定窗口的参数设定按钮。进入

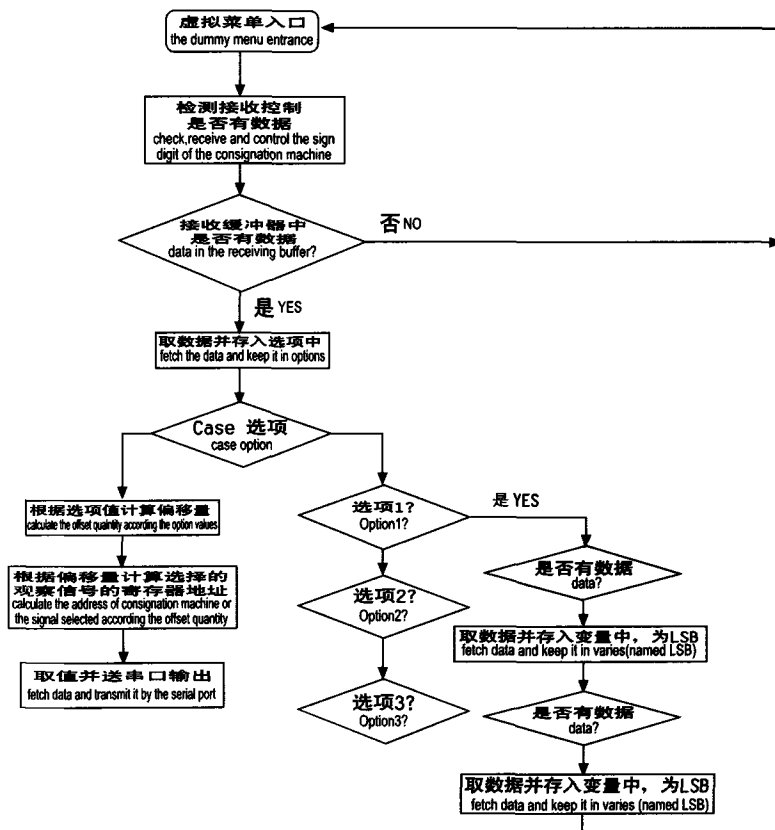


图 4 DSP 虚拟菜单及输出控制图

Figure4. The frame of DSP Dummy Menu and Output Control

data word.

5. The Design of DSP Communication

The programmable SCI module of TMS320F240 sustains numeral communication between CPU of against returning zero format and other synchronisms outside equipments. The module mostly includes two data buffer (receiving buffer and sending buffer) with absolute enable bit and shielded bit, programmable baud rate befalls, mapping control register in the data section and state register. The buffer data received and sent by SCI is 8 bit. The enable bit of RX、TX、clk is controlled by SCI control register 1, and the shield of SCI interruption is controlled by SCI control register2. The baud rate is enacted by SCIHBAUD and SCILBAUD.

The I/O communication of SCI receiving and sending port is controlled by SCI port control register. These settings are finished in the initialization of SCI.

SCI initialization

```
LDP DP_PFI
SPLK #00010111b, SCICCR ; one
stop bit ,no partity, 8 bits
SPLK #0013h, SCICTL1 ; enable RX,
TX,clk
SPLK #0000h, SCICTL2 ; disable
SCI interrupts
SPLK #0000h, SCIHBAUD ; MSB
*****
SPLK #0040h, SCILBAUD : LSB
9600 Baud for sysclk 10Mhz
SPLK #0022h, SCIPC2 : I/O setting
SPLK #0033h, SCICTL1 ; end ini-
tialization
```

SCI initialization

LDP DP_PFI
SPLK #00010111b, SCICCR ; one
stop bit ,no partity, 8 bits
SPLK #0013h, SCICTL1 ; enable RX,
TX,clk
SPLK #0000h, SCICTL2 ; disable SCI
interrupts
SPLK #0000h, SCIHBAUD ; MSB
SPLK #0040h, SCILBAUD : LSB
9600 Baud for sysclk 10Mhz

观察值给定分支后,我们设计了用控制字计算观察信号地址偏移量的方法,计算出的偏移量可以直接用于输出寄存器的选择,使程序得到大大的简化。在进入参数设定分支后,程序通过判断,依据协议读入下一个或两个传输字作为数据字。

串行口的输入和输出控制相对独立,既实现了双工传输,又可以进行实时控制。

6、注意事项及结论

在以上述方法实现 F240 与 PC 机通信时,应注意一下事项:

1、速度信息的上传可通过两种方式进行: 传光码盘的脉冲数或直接传转速。而转速和光码盘的脉冲数通常都大于 256, 所以,无论是直接上传转速,还是上传光码盘的脉冲数,在上位机的计算转速中,速度信息都要两个字节才能表示。

2、电流信息通常是指 A/D 转换后得到的定子电流信息,但也可以是直接转矩控制中经过坐标变换后得到的电流信息。这些量通常是小数,在传输前通常要作一定的处理,本文用到的方法是乘上一个倍数后取整数,上传到上位机后在除以相同的倍数。为保证传输的精度,同时考虑到不能使传输的数据量增加太多,本文中此倍数取 1000。显然,电流量也需要两个字节表示。

在结合微机部分和 DSP 部分的通信程序后,系统初步实现了基于双工串行同的 F240 内部变量的监测和控制。

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SPLK #0022h, SCIPC2 : I/O setting
SPLK #0033h, SCICTL1 : end initialization

The structure of communication program of DSP and PC part program are corresponding one by one. Therefore it has many possible exports, such as the choosing of observation variables, the setting of the functions, the presenting of rate, and so on.

In order to optimize the assembler and reduce the unsteadiness made by multi-relect and program interruption, we adopt the dummy menu, shown as figure4:

The input control of dummy menu is the transfer word of serial port. Receiving one transfer word, the dummy menu judges firstly whether it belongs to command of observation variable given or function set. The former is corresponding to button option of main window, and the other is corresponding to setting parameter button of setting parameter window. Enter the embranchment of observation variable given; we design the way of using control word to calculate the address offset of observation signal. The offset calculated can be used directly on the choosing of the output register. In the way, the program is simplified greatly

Enter the embranchment of setting parameter, through judgment the program reads the next one or two transfer words as data according to the protocol.

In order to reduce the influence of interruption made by a mass of jump in dummy menu, we put the judge program on the options into main program, while put the serial output module into interruption program whose control is finished by the change of register offset. Therefore the input and output control of the DSP serial port is independent relatively, which not only achieves duplex transfers, but also carries through real time control.

6. Notices and Conclusion

The proceedings to be paid attention to and conclusions

There are some proceedings to be paid attention to when the communication between F240 and PC is carried out by the means motioned above.

1. There are two means to transmit above the speed message, which are the transmission of the pulse number of the compact disk and the transmission of the revs. The pulse number of the revs and the compact disk commonly is larger than 256. Therefore, the speed message needs two bytes to be expressed no matter what means should be by.

2. The current message generally means the current message of stator which is get after AD conversion, but it can also means the current message which is get after coordinate conversion in the direct torque control. These quantities generally are decimal fraction, which need to be deal with before transmission.

The way used in the article is to multiply multiple and get the integer. Then the number is transmit above to the epigyny machine and divides the same multiple. Consider the transmission precision and the increased data quantity; the multiple in this article is adopted as 1000. Obviously, the current quantity also needs two bytes to be expressed.

After the communication program combing the microcomputer part and the DSP part is carried out, the system elementarily carries out the inspection and control of the inner variables of F240 according the bi-directional serial port and collateral port.

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